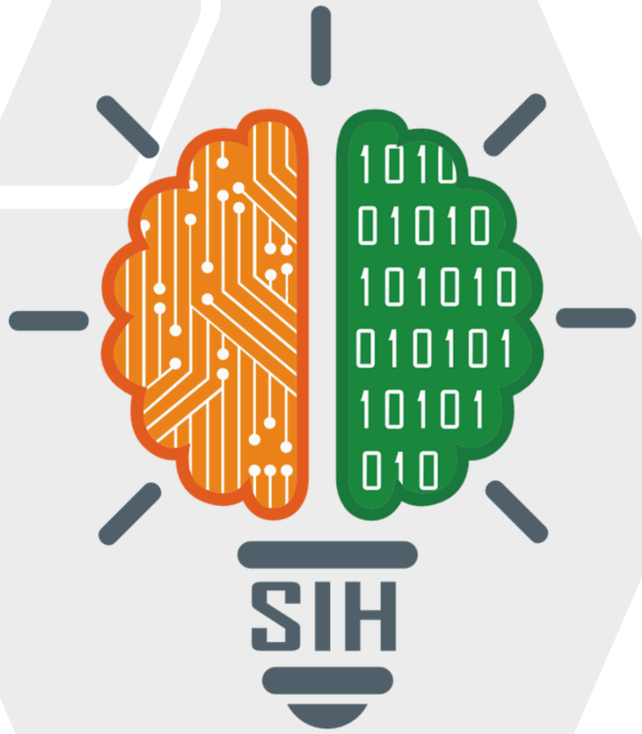
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- **Problem Statement ID –** 26068
- **Problem Statement Title-** WeatherGPT: Conventional AI for Weather Forecasting Alerts, and Climate information.
- **Theme-** Disaster Management
- **PS Category-** Software
- **Team ID-** 149059
- **Team Name (Registered on portal)-** VisionX123

VisionX123

MausamSetu

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• PROPOSED SOLUTION

MausamSetu
Weather • Risk • Action

Idea / Solution / Prototype

- **MausamSetu** combines forecasts, alerts, satellite and climate data in one platform.
- Users can ask weather-related questions through text or voice.
- It analyzes the user's location, query and weather conditions.
- Provides **risk level + clear actions** (e.g., travel, avoid risky areas, take precautions).
- Prototype includes a weather dashboard, risk analysis, alerts and chatbot interface.

Innovation & Uniqueness

- **Weather → Risk → Action:** Converts complex information into practical decisions.
- **Hyperlocal Intelligence:** Location-based risks and recommendations.
- **Conversational AI:** Natural language queries instead of multiple portals.
- **Unified Weather Data:** Combines forecasts, warnings, satellite and climate data.
- **Inclusive Access:** Supports voice, regional languages and visual alerts.



Multi-language
Voice Support
Location Based

Solution Approach

- **Collect:** Get data from IMD, satellite, radar and climate sources.
- **Understand:** AI processes the user's query, location, time and intent.
- **Analyze:** Weather engine studies current and forecast conditions.
- **Assess:** Identifies risks like heavy rain, floods, cyclones and extreme weather.
- **Recommend:** Generates simple, situation-specific actions.
- **Deliver:** Provides results through chat, voice, alerts and visual information.



Smarter Weather. Safer Tomorrow.

What is Different from Other Weather Platforms?

- Gives what to do, not just weather information.
- Provides location-specific risks and recommendations.
- Combines multiple weather sources in one platform.
- Supports natural-language, voice and regional languages.
- Offers guidance for citizens, farmers, travelers and authorities.



PROTOTYPE

LINK

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TECHNICAL APPROACH

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Technology Used

Programming Language

- **Python** – AI, NLP & backend
- **JavaScript** – Interactive frontend
- **HTML/CSS** – Web interface

Frameworks & Libraries

- **FastAPI/Flask** – Backend API
- **React.js** – Frontend
- **LLM/NLP APIs** – AI responses
- **Pandas/NumPy** – Data processing
- **Leaflet/Map APIs** – Maps & risk visualization

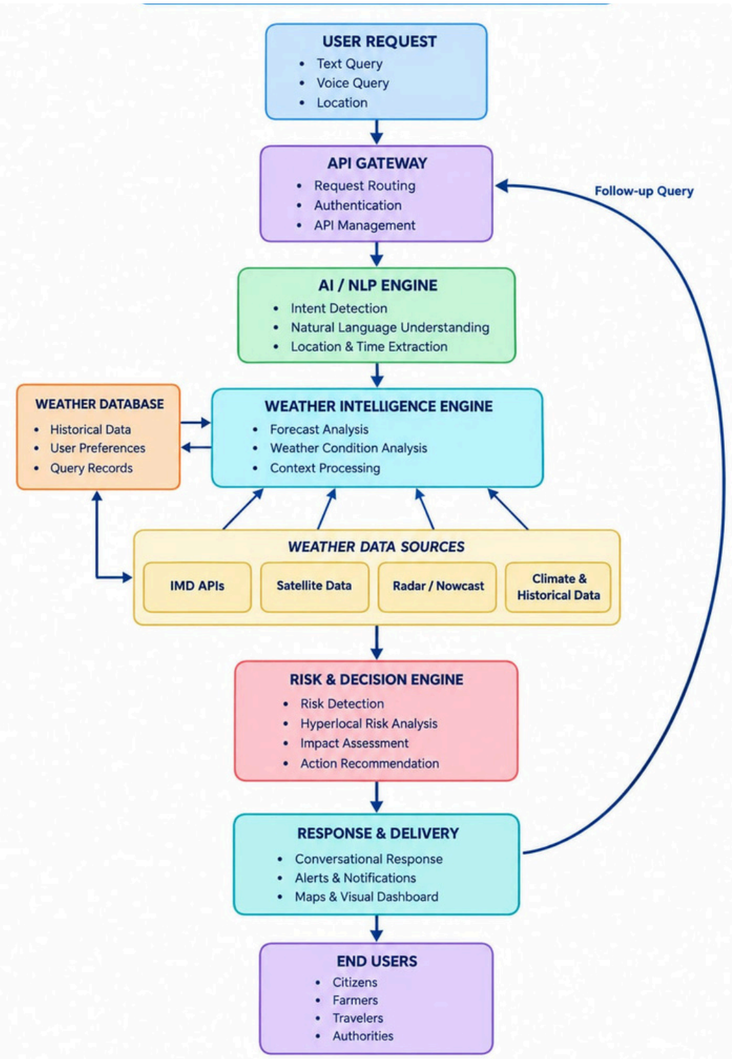
Data & APIs

- **IMD APIs** – Forecasts & alerts
- **Satellite/Radar Data** – Weather monitoring
- **Climate Data** – Historical analysis

Database & Development

- **PostgreSQL/MongoDB** – Data storage
- **Cloud Platform** – Deployment

SYSTEM ARCHITECTURE



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FEASIBILITY AND VIABILITY

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FEASIBILITY

API-First Architecture
Built on existing meteorological data sources (IMD, NOAA, ISRO satellite feeds) via APIs, so no need to build forecasting models from scratch — only the conversational and alerting layer needs original development.

Mature NLP Tooling
Open-source LLM frameworks and speech-to-text libraries let the conversational interface be prototyped quickly without proprietary AI infrastructure.

Cloud-Native Deployment
Runs as a lightweight web/app/chatbot service, so no specialized hardware or field installation is required, keeping setup cost and deployment time low.

Incremental Rollout
Core modules (query answering, alert push, climate Q&A) can be built and tested independently, then merged — reducing integration risk before the final demo.

VIABILITY

Rising Real-World Need
Increasing frequency of extreme weather events (heatwaves, floods, cyclones) creates genuine, growing demand for accessible, plain-language weather guidance.

Multilingual & Voice Access
Regional-language and voice support extend reach to rural and low-literacy users who can't easily interpret technical weather bulletins today.

Trusted Data Backbone
Sourcing directly from government meteorological data lends the tool credibility that a purely third-party weather app wouldn't have.

Established User Comfort
Widespread adoption of voice assistants and chatbots shows people are already comfortable asking natural-language questions to AI — lowering the adoption barrier.

ECONOMICALLY FEASIBLE

Low Infrastructure Cost
Uses existing cloud infrastructure and APIs.

No Special Hardware
Works with standard web and cloud systems.

Open-Source Tools
Uses cost-effective AI, NLP and development libraries.

Scalable
Resources can increase gradually as users grow.

Reusable Data Sources
Integrates existing weather APIs instead of building costly systems from scratch.

Long-Term Value
Helps reduce losses by enabling early warnings and better decisions.

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IMPACT AND BENEFITS

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IMPACT

Faster Response
Early alerts enable quick action during extreme weather.

Safer Communities
Hyperlocal warnings help people avoid risky areas.

Reduced Losses
Timely forecasts reduce crop, travel and economic losses.

Better Preparedness
Early warnings improve disaster readiness.

BENEFITS

One-Stop Platform
Forecasts, alerts and climate data in one place.

Actionable Advice
Tells users what to do, not just the forecast.

Inclusive Access
Multilingual and voice-based weather assistance.

Smarter Decisions
Supports citizens and authorities with relevant insights.

Key Risks & Solutions

KEY RISK	SOLUTION
DATA ACCURACY	Use verified official weather sources and data validation.
AI MISUNDERSTANDING	Query validation and fallback responses for unclear requests.
EMERGENCY DELAYS	Priority processing for critical weather alerts.
MULTI-LANGUAGE SUPPORT	Phased rollout with regional language validation.
HIGH API / DATA LOAD	Scalable cloud infrastructure and efficient data processing.
FALSE OR UNCERTAIN PREDICTIONS	Cross-check multiple reliable data sources before generating responses.

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RESEARCH AND REFERENCES

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IMD

- Weather forecasts and official alerts.

ISRO MOSDAC

- Satellite weather data.

WMO

- Early-warning systems and disaster management.

GraphCast Research Paper

- AI-based global weather forecasting.

Pangu-Weather Research Paper

- Deep-learning-based weather prediction.

Research Focus

- Conversational AI, hyperlocal risk and actionable weather alerts.

UI/UX

